

Factorial of a Number — Recursive and Iterative

Question: Find the factorial of a number using recursive and iterative methods. Compare their execution time and determine their time and space complexities.

Test input: $n = 10$. The calculation is repeated 1,000,000 times to make the execution-time comparison measurable.

C Program

```
#include <stdio.h>
#include <time.h>

#define REPEAT 1000000

unsigned long long factorialRecursive(int n) {
    if (n <= 1)
        return 1;
    return n * factorialRecursive(n - 1);
}

unsigned long long factorialIterative(int n) {
    unsigned long long fact = 1;
    for (int i = 2; i <= n; i++)
        fact *= i;
    return fact;
}

int main() {
    int n;
    printf("Enter a number: ");
    scanf("%d", &n);

    if (n < 0) {
        printf("Factorial is not defined for negative numbers.\n");
        return 0;
    }

    unsigned long long recursiveResult = 0, iterativeResult = 0;
    clock_t start, end;

    start = clock();
    for (int i = 0; i < REPEAT; i++)
        recursiveResult = factorialRecursive(n);
    end = clock();
    double recursiveTime =
        (double)(end - start) * 1000.0 / CLOCKS_PER_SEC;

    start = clock();
    for (int i = 0; i < REPEAT; i++)
        iterativeResult = factorialIterative(n);
    end = clock();
    double iterativeTime =
        (double)(end - start) * 1000.0 / CLOCKS_PER_SEC;

    printf("\nFactorial of %d = %llu\n", n, recursiveResult);
    printf("Recursive Method : %.3f ms\n", recursiveTime);
    printf("Iterative Method : %.3f ms\n", iterativeTime);

    printf("\nTime Complexity:\n");
    printf("Recursive : O(n)\n");
    printf("Iterative : O(n)\n");

    printf("\nSpace Complexity:\n");
    printf("Recursive : O(n) due to function call stack\n");
    printf("Iterative : O(1)\n");

    return 0;
}
```

Actual Output

Enter a number:
Factorial of 10 = 3628800
Recursive Method : 22.239 ms
Iterative Method : 10.807 ms

Time Complexity:
Recursive : $O(n)$
Iterative : $O(n)$

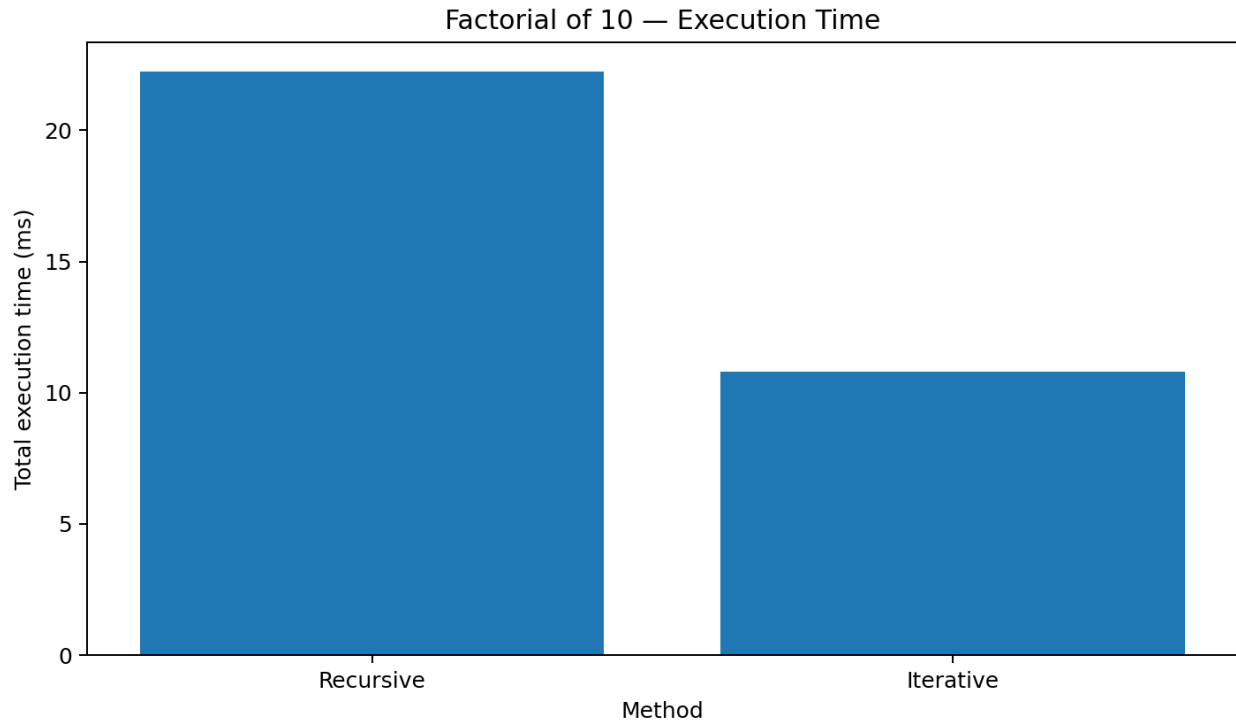
Space Complexity:
Recursive : $O(n)$ due to function call stack
Iterative : $O(1)$

Complexity Analysis

Method	Time Complexity	Space Complexity
Recursive Factorial	$O(n)$	$O(n)$ — recursion stack
Iterative Factorial	$O(n)$	$O(1)$

Conclusion: Both methods have $O(n)$ time complexity. Recursive factorial uses $O(n)$ extra stack space, whereas the iterative method uses $O(1)$ extra space. Therefore, the iterative method is more space-efficient.

Graph — Execution Time Comparison



The graph was generated using Python Matplotlib from the actual timing values produced by the C program.